13. Scenario: You are working on a text analysis project and need to determine the frequency distribution of words in a given text document. You have a text document named "sample_text.txt"
containing a paragraph of text. Your task is to develop a Python program that reads the text document, processes the text, and generates a frequency distribution of the words.
Question: How would you develop a Python program to calculate the frequency distribution of words in a text document?

```
import re
from collections import Counter
with open("sample_text.txt", "w") as file:
    file.write("Python is easy. Python is powerful.")
with open("sample_text.txt", "r") as file:
    text = file.read()
text = text.lower()
words = re.findall(r'\b[a-z]+\b', text)
frequency = Counter(words)
print("Word Frequency Distribution:")
for word, count in frequency.items():
    print(word, ":", count)
```

```
Word Frequency Distribution:
python : 2
is : 2
easy : 1
powerful : 1
```

14. Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data
frame.
Question: Develop a code in python to find the frequency distribution of the ages of the customers
who have made a purchase in the past month.

```
import pandas as pd
data = {
    'Age': [25, 30, 25, 35, 30, 40, 25, 35, 30, 25]
}
df = pd.DataFrame(data)
frequency = df['Age'].value_counts().sort_index()
print("Frequency Distribution of Customer Ages:")
print(frequency)
```

```
Frequency Distribution of Customer Ages:
Age
25    4
30    3
35    2
40    1
Name: count, dtype: int64
```

15. Scenario: You are a data analyst working for a social media platform. As part of your analysis,
you have a dataset containing user interaction data, including the number of likes received by each
post. Your task is to develop a Python program that calculates the frequency distribution of likes among the posts.
Question: Develop a Python program to calculate the frequency distribution of likes among the posts?

```
import pandas as pd
data = {
    'Likes': [10, 20, 10, 30, 20, 10, 40, 30, 20, 10]
}
df = pd.DataFrame(data)
frequency = df['Likes'].value_counts().sort_index()
print("Frequency Distribution of Likes:")
print(frequency)
```

```
Frequency Distribution of Likes:
Likes
10    4
20    3
30    2
40    1
Name: count, dtype: int64
```

16. Scenario: You are working on a project that involves analyzing customer reviews for a product.
You have a dataset containing customer reviews, and your task is to develop a Python program that
calculates the frequency distribution of words in the reviews.
Question: Develop a Python program to calculate the frequency distribution of words in the customer reviews dataset?

```
import pandas as pd
import re
from collections import Counter
data = {'Reviews': ['Product is good','Product is excellent',
        'Good product','Excellent quality']}
df = pd.DataFrame(data)
text = ' '.join(df['Reviews']).lower()
words = re.findall(r'\b[a-z]+\b', text)
frequency = Counter(words)
print("Frequency Distribution of Words:")
for word, count in sorted(frequency.items()):
    print(word, ":", count)
```

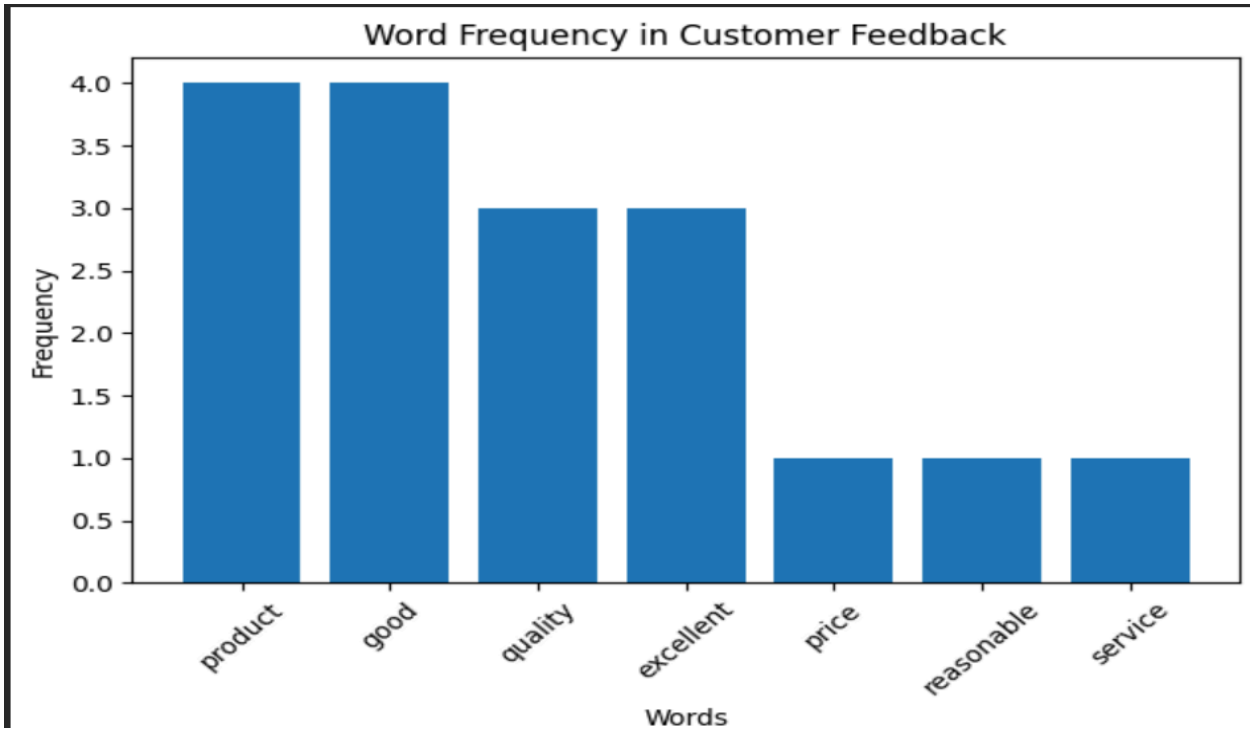
```
Frequency Distribution of Words:
excellent : 2
good : 2
is : 2
product : 3
quality : 1
```

17. Scenario: You are a data analyst working for a marketing research company. Your team has collected a large dataset containing customer feedback from various social media platforms. The dataset consists of thousands of text entries, and your task is to develop a Python program to analyze the frequency distribution of words in this dataset. Your program should be able to perform the following tasks: Load the dataset from a CSV file (data.csv) containing a single column named "feedback" with each row representing a customer comment. Preprocess the text data by removing punctuation, converting all text to lowercase, and eliminating any stop words (common words like "the," "and," "is," etc. that don't carry significant meaning).Calculate the frequency distribution of words in the preprocessed dataset. Display the top N most frequent words and their corresponding frequencies, where N is provided as user input. Plot a bar graph to visualize the top N most frequent words and their frequencies. Question: Create a Python program that fulfills these requirements and helps your team gain insights from the customer feedback data.

```python
import pandas as pd
import re
import matplotlib.pyplot as plt
from collections import Counter
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS
data = {"feedback": ["The product is good and the quality is excellent",
"The product is very good","Quality is good and price is reasonable",
"The product quality is excellent","Good product and excellent service"]}
df = pd.DataFrame(data)
df.to_csv("data.csv", index=False)
df = pd.read_csv("data.csv")
stop_words = set(ENGLISH_STOP_WORDS)
words = []
for feedback in df["feedback"].dropna():
    feedback = feedback.lower()
    feedback = re.sub(r'[^a-z\s]', '', feedback)
    for word in feedback.split():
        if word not in stop_words:
            words.append(word)
frequency = Counter(words)
N = int(input("Enter the number of top words to display: "))
top_words = frequency.most_common(N)
print("Top", N, "Most Frequent Words:")
for word, count in top_words:
    print(word, ":", count)
if top_words:
```

```python
    words_list, counts = zip(*top_words)
    plt.bar(words_list, counts)
    plt.xlabel("Words")
    plt.ylabel("Frequency")
    plt.title("Word Frequency in Customer Feedback")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

```
Enter the number of top words to display: 8
Top 8 Most Frequent Words:
product : 4
good : 4
quality : 3
excellent : 3
price : 1
reasonable : 1
service : 1
```



18. Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result.

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

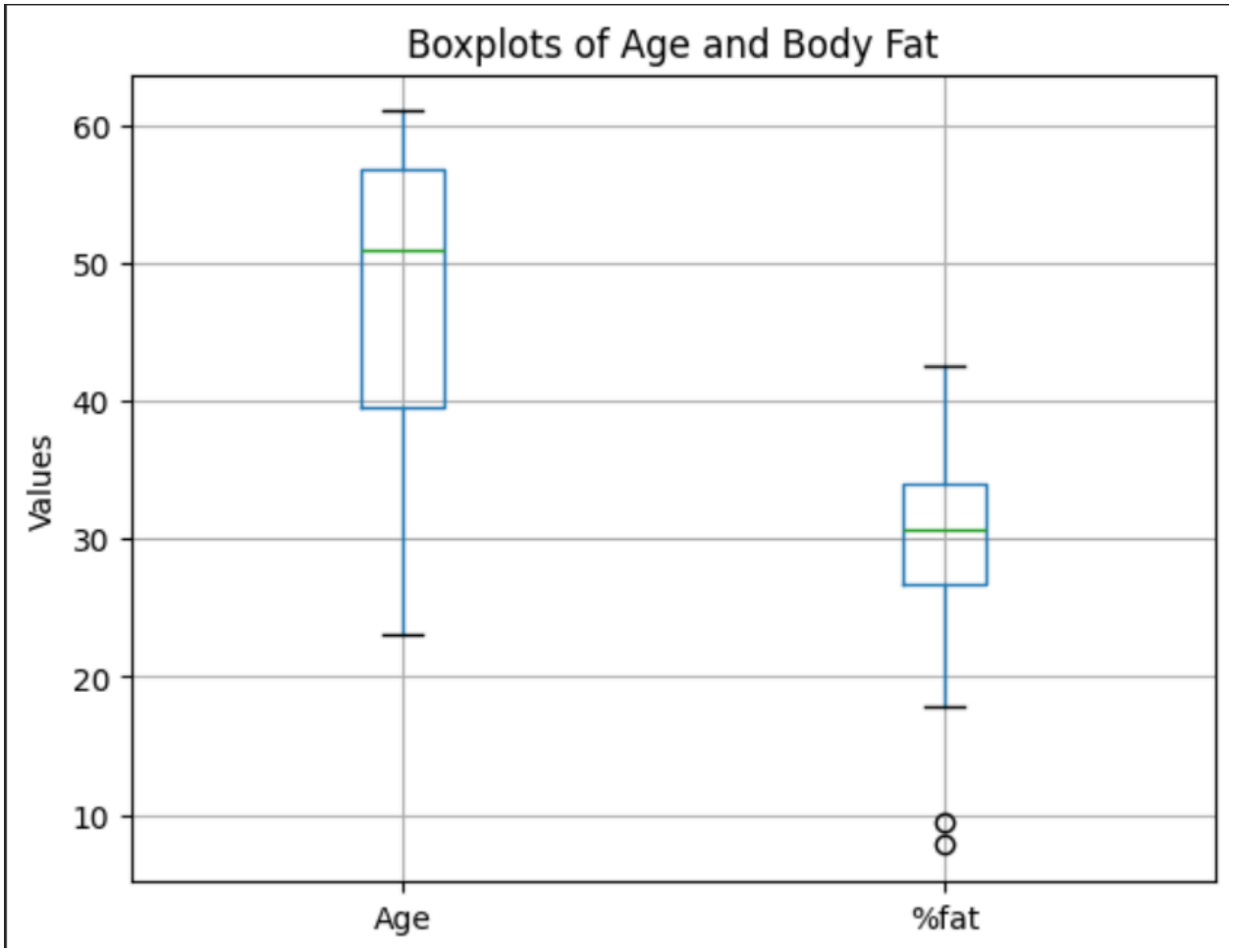
Question: 🎬 Calculate the mean, median and standard deviation of age and %fat using Pandas. 🎬 Draw the boxplots for age and %fat.

```python
import pandas as pd
import matplotlib.pyplot as plt
data = {'Age': [23,23,27,27,39,41,47,49,
50,52,54,54,56,57,58,58,60,61],
'%fat': [9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,
31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7]}
df = pd.DataFrame(data)
print("Mean:")
print(df.mean())
print("\nMedian:")
print(df.median())
print("\nStandard Deviation:")
print(df.std())
df.boxplot(column=['Age', '%fat'])
plt.title("Boxplots of Age and Body Fat")
plt.ylabel("Values")
plt.show()
```

```
Mean:
Age     46.444444
%fat    28.783333
dtype: float64

Median:
Age     51.0
%fat    30.7
dtype: float64

Standard Deviation:
Age     13.218624
%fat     9.254395
dtype: float64
```


Boxplots of Age and Body Fat
Values
60
50
40
30
20
10
Age
%fat